Memorandum

To: Prof. Daniel, Chair of the Smith School of Chemical and Biomolecular Engineering

From: Ben Mitlitski, Charlotte Larson, Ian Vann, and Loobna Shego

Date: 17 December 2025

Subject: Design Process for Olin Hall Thermal Battery

EXECUTIVE SUMMARY

The purpose of this study is to design a thermal battery system to store and release thermal energy from the waste heat from Olin Hall's steam distribution line. This system supports Cornell's Climate Action Plan and its goal of carbon neutrality by 2035 by reducing fossil fuel use and enhancing energy efficiency. An experimental analysis of a heat exchanger currently in the Olin Hall Unit Operations Lab (HX3) was conducted to characterize heat transfer under varying flow conditions and inform the design of our thermal battery system. Our design consists of a hexagonal tank, densely packed with industrial-grade barium hydroxide octahydrate ($Ba(OH)_2 \cdot 8H_2O$) as the phase change material (PCM) used to store energy as latent heat. We propose a design consisting of two concentric copper coils. The inner coil will contain the hot exhaust steam, and the outer coil will contain the cold water to which energy will eventually be discharged. After passively charging throughout an 18-hour period, the proposed design is capable of heating cold water (15°C, 1.2 kg/s) to 61.5°C for 7.4 hours per day, accounting for a 1.4-hour period of overlap. This analysis indicates that a PCM-based thermal battery is a feasible and impactful near-term solution for reducing energy waste in Olin Hall. Calculations confirm that the battery is capable of storing 6.39 GJ of energy, saves 66,563 kg CO_2 per year, and the benefits and cost savings of the system outweigh the capital cost of 27,000 USD, which supports the implementation of this thermal battery system.

TABLE OF CONTENTS

EXECUTIVE SUMMARY	1
INTRODUCTION	3
PROCESS AND PERFORMANCE ANALYSIS OF HX3	3
DESIGN PROPOSAL OF THERMAL BATTERY FOR OLIN HALL	5
INTEGRATION WITH CLIMATE ACTION PLAN	11
RECOMMENDATIONS AND CONCLUSIONS	12
APPENDICES	12
REFERENCES	17

INTRODUCTION

The purpose of this study is to design and implement a thermal battery system in Olin Hall charged via the steam distribution exhaust. The existing infrastructure of Olin Hall has a portion of hot steam exiting the building at approximately 150ºC and 3 bar. While there is no inherent problem with this exhaust stream, it contains a significant amount of recoverable thermal energy.

A traditional heat exchanger transfers heat across a temperature gradient via convection and conduction. The Olin Hall exhaust is a hot stream of constant fluid flow rate but there is not a constant demand for heating. As a result, we concluded that the implementation of a thermal battery would be advantageous for several reasons, including operating and implementation costs, efficiency, and reduction of emissions. We believe a thermal battery in Olin Hall will reduce fossil fuel use and improve energy efficiency.

Thermal batteries store energy as it becomes available and release it when there is a demand. This technology is critical for the renewable energy space. Storing energy in a thermal medium, whether it be a fluid tank, carbon-based solid, or PCM, allows for large-scale energy storage and release. We believe that the current exhaust stream provides a similar opportunity to renewables, allowing for charging and discharging of a thermal battery to provide heating to Olin Hall. This memo discusses our proposal for an efficient thermal battery that could be implemented in Olin Hall to support Cornell's focus on sustainability.

PROCESS AND PERFORMANCE ANALYSIS OF HX3

To inform our thermal battery design, we analyzed Cornell's Heat Exchanger 3 (HX3). HX3 employs concurrent tube-and-shell flow to transfer heat between hot and cold water.

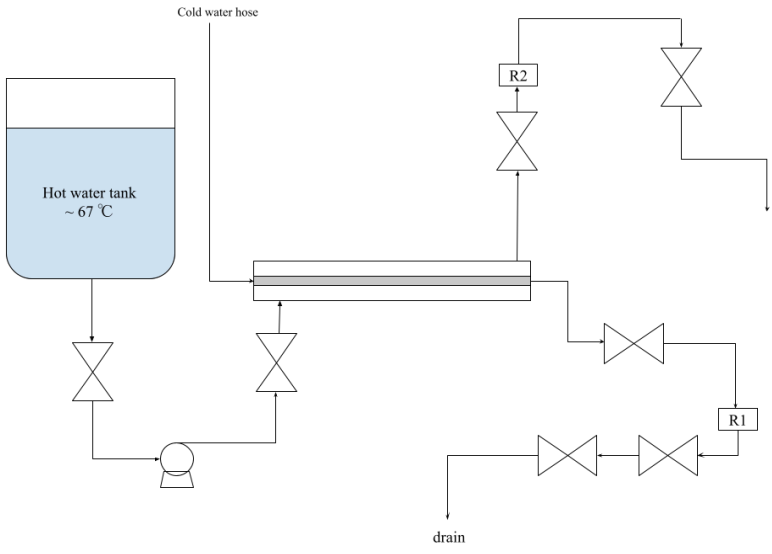


Figure 1. Process Flow Diagram of HX3

As demonstrated in Figure 1 above, cold water flows in a tube while hot water flows in the surrounding shell, both entering the heat exchanger on the same side. The rotameters for the cold and hot streams, labeled R1 and R2, respectively, allow for the inlet flow rates of each stream to be adjusted. In adjusting flow rates and measuring inlet and outlet temperatures for each stream, we were able to relate the temperature change, and consequently the quantity of energy transferred per unit time, to the stream flow rates. Knowing these values as well as significant thermodynamic properties, such as heat capacity, we calculated the tube-side (inner) and shell-side (outer) convective heat transfer coefficient (h_i and h_o), total heat transfer coefficient (U), and Reynolds number for each tested combination of flow rates. The Reynolds number calculated from all trials ranged from 3,835 to 23,010, confirming that HX3 operates entirely in the turbulent regime, where convection dominates heat transfer.

We first used the heat transfer rate (Eq. 1) to determine the overall heat transfer coefficient (Eq. 2). The value of the overall heat transfer coefficient ranged from 1,350 to 3,410 W/m^2K, and had an average value of 2,270 W/m^2K. To determine the heat transfer coefficients, we conducted two experiments: one in which the flow rate of the cold fluid was held constant and that of the hot fluid varied, and vice versa. By generating Wilson plots (Eq. A5) with the Sieder-Tate modification (Eq. A7, Graph 1) and the Donohue equation (Eq. A8, Graph 2), we graphically determined the inner and outer convective heat transfer coefficients of HX3. The values of the inner convective heat transfer coefficient ranged from 1,950 to 8,890 W/m^2K and had an average value of 5,100 W/m^2K (Eq. A6). The outer convective heat transfer coefficient ranged from 923 to 12,700 W/m^2K, and had an average value of 4,590 W/m^2K (Eq. A9). The convective heat transfer coefficients generally increase with velocity.

The analysis of HX3 informed our desired fluid flow rates and heat transfer coefficients. From experimenting with the flow rate of the hot stream in HX3, we found that heat transfer from liquid water was maximized at a flow rate of approximately 20 gal/min (1.4 m/s). We assume that the optimal flow rates for heat transfer in a horizontal concurrent heat exchanger correspond to efficient heat transfer from fluid flow in our thermal battery. Because our proposed design incorporates hot water, we decided to set our tube dimensions such that water flows at around 1.4 m/s, achieving near maximum heat transfer to the storage material in our thermal battery. We followed similar logic for the flow rate of cold water in HX3 and our thermal battery system. In HX3, heat transfer to the cold stream, after experimenting with different flow rates, was found to be maximized at 19.5 gal/min (4.3 m/s). We make the same assumption as previously and use this flow rate in our design.

Our design, as is detailed later in this memo, involves coils in a hexagonal prism packed with a PCM. The hot and cold streams infrequently flow at the same time, and are never in direct contact with one another.

On the contrary, HX3 involves a simple tube-and-shell arrangement with concurrent flow of hot and cold water. While the hot side of our thermal battery was deemed to be too different from HX3 to be applicable, we were able to apply the overall heat transfer coefficient to our cold stream. Utilizing the same water inlet as HX3 (15°C), the same tube dimensions, and an analyzed flow rate, we were able to utilize an overall heat transfer coefficient of 2,377 W/m^2K for flow at 19.5 gal/min. We used the maximum observed heat transfer from HX3 to model heat transfer to the cold tube in our thermal battery, as our goal is to supply heat to the cold stream and raise its temperature as efficiently and as much as possible.

DESIGN PROPOSAL OF THERMAL BATTERY FOR OLIN HALL

The current Olin Hall steam distribution line contains an exhaust stream with a flow rate proportional to the load of the building. The exhaust is steam at 3 bar at 150°C. We assume exhaust steam leaves at 0.075 kg/s during standard operational hours of Olin Hall (6:00 AM - 8:00 PM daily). During hours of reduced operation (8:00 PM-6:00 AM), we assume the mass flow rate of the steam is 0.015 kg/s (20% of standard operational load). Currently, this steam is released to the atmosphere, leaving potential for further heat extraction.

The saturation temperature of steam at 3 bar is 133.5°C[1], a 16.5°C temperature drop from the provided exhaust stream. Steam is generally easier to transport and dispose of than liquid water, as gas can rise naturally without the need for any additional pumping. Although we considered a design maintaining an outlet exhaust temperature of 135°C (slightly above the saturation temperature to ensure an entirely vapor stream), the energy extracted by the thermal battery in this configuration was only 1.7% of that achievable under condensing conditions (Eq. B14). We acknowledge that involving condensation in the heat exchanger introduces new complications with regard to fluid flow, rates of convective heat transfer, and disposal. However, to achieve optimal heat transfer and storage of energy for further use, we ultimately decided that the exhaust stream would need to be condensed to a liquid.

We elected to use barium hydroxide octahydrate ($Ba(OH)_2 \cdot 8H_2O$), a Phase Change Material (PCM), for thermal energy storage over alternative options, such as a simple water tank or utilizing a carbon-based option, such as bricks. We determined that, to hold the daily charging capacity, this PCM requires 60% less volume than bricks or water tanks (Eq. B1-B3), proving much more feasible considering spatial limitations in Olin Hall. The brick-based heat storage material allows for a significantly lower mass and potential to have a smaller footprint, but it requires intense heating of bricks, which severely limits the

[1] Koretsky, M. D. Engineering and Chemical Thermodynamics., 2nd ed.; John Wiley: S.L., 2013.

practicality of this option, especially when accounting for cooling infrastructure and safety. For these reasons, this storage material was excluded. We concluded that this PCM provides the optimal latent heat to volume ratio, while maintaining relatively low temperatures (less than 150°C) and mitigating safety concerns.

At steady-state conditions, the exhaust stream will transfer heat to the PCM at the material's melting point. Similarly, when discharging, heat will transfer from the PCM to the cold stream at the same temperature. The goal of this thermal battery is to maximize the heat transfer rate between the PCM and both hot and cold streams.

The PCM needs to have a melting point between the temperature of the inlet cold water for building heating (15°C) and the outlet exhaust stream (150°C). At operating conditions, we assume the PCM to always remain at its melting point. Therefore, all charging and discharging of the system is supplied or retrieved as latent heat in the melting and solidifying of the PCM, rather than sensible heat. To enable effective PCM charging from the hot exhaust stream, this melting point must be considerably below 150°C and, ideally, below 100°C, permitting eventual condensation to liquid water at atmospheric pressure. Moreover, to be able to effectively discharge heat to the 15°C cold stream, the selected PCM needs to have a melting point considerably above 15°C.

We determined that an effective PCM for our thermal battery would be barium hydroxide octahydrate ($Ba(OH)_2 \cdot 8H_2O$). This salt was selected for its high latent heat to volume ratio, affordability, and ideal melting point (~78°C).[2]

The PCM is scored in a standard, insulated 8.5 m^3 tank. This tank can be densely packed with 22,900 kg of industrial-grade barium hydroxide octahydrate (Eq. B1), which costs approximately 12,600 USD (Eq. D1).[3] Our suggested system contains a series of two tubes: an outer and an inner coil. The outer coil (detailed later) contains the cold water for energy discharge. The exhaust steam flows through the inner coil, traversing the tank in a helical path, allowing for a longer tube length in a set volume of the PCM tank. This tube is completely surrounded by PCM, ensuring maximum surface area for heat transfer. We considered adding external fins to this coil, but ultimately concluded that these fins would merely increase surface area for conduction, as the convective heat transfer coefficient for the hot steam inside the coil is expected to be significantly greater than that of the PCM outside. Still, this would slightly improve heat

[2] Martínez, F. R.; Borri, E.; Kala, S. M.; Ushak, S.; Cabeza, L. F. Phase change materials for thermal energy storage in industrial applications. *Heliyon* **2024**, *11*, e41025.
[3] CAS: 17194-00-2 Factory Price Barium Hydroxide Octahydrate 99% Ba (OH) 2.8H2O. Made-in-China.com. https://gainer.en.made-in-china.com/product/JnBYOUKyYDRA/China-CAS-17194-00-2-Factory-Price-Barium-Hydroxide-Octahydrate-99-Ba-OH-2-8H2O.html (accessed 2025-12-17).

transfer efficiency, but we elected to exclude fins on this coil because we did not believe that they would be worth the added cost.

Rather than arbitrarily setting our process space and then quantifying its resulting heat transfer rate and efficiency, we chose to define heat transfer stages and design our thermal battery to obtain the desired results. The exhaust stream is ultimately converted to hot water, so there is no true motivation to keep the stream in a vapor state throughout the battery. Similarly, a mixture of liquid water and vapor is much more difficult to transport and dispose of, primarily due to our inability to pump water droplets. As a result, we ensured that our design allowed for enough energy transfer to completely condense the incoming steam in the thermal battery. We decided to make our outlet stream water at 95°C, which is below its saturation temperature of 100°C at 1 bar.

To achieve this, we can model the heat extraction process as four distinct stages. Each stage has a specified temperature and pressure of the stream (Table 1).

Table 1. Outline of the Stages of Fluid Flow Through Our Proposed Thermal Battery

Process Stage	Temperature (°C)	Pressure (bar)	Phase	Mass Flow Rate (kg/s)
1	150	3	vapor	0.075
2	133.5	3	vapor	0.075
3	133.5	3	liquid	0.075
4	95	3	liquid	0.075
5	95	1	liquid	0.075

To summarize, we will:

1) Cool the incoming steam to the saturation temperature for steam at 3 bar.
2) Condense all of the steam from vapor to liquid isothermally at the saturation temperature.
3) Cool the liquid water from the saturation temperature to 95°C.
4) Expand the hot water from 3 bar to 1 bar.

To maximize heat transfer from the hot exhaust stream to the PCM, we must ensure that the linear velocity of the steam is within a reasonable range. For example, it is entirely possible to design a system that transfers steam at nearly 400 m/s, but it is highly probable that steam moving slower, and consequently spending more time in the heat exchanger, would transfer more heat to the PCM, thus improving the device's efficiency. We determined that it is reasonable to have steam flowing through the

system at 30-40 m/s,[4] as this is an achievable intersection of design feasibility and effectiveness. We calculated that increasing the inner diameter of the tube to 40 mm would change the flow velocity to our desired value (Eq. B4). Similarly, based on our analysis of HX3, we understood that heat transfer from a liquid is maximized when the liquid flows at approximately 1.4 m/s. From this, we designed our tubing to shrink down to an inner diameter of 10 mm once the steam condenses into liquid water (Eq. B5). To determine at which point along the tube the steam is fully condensed, we use standard heat transfer correlations to determine the time it takes for our mass of steam to condense, and, knowing the steam's time to condense and velocity, we can determine the length of tube required (Eq. B10). These changes allow for more effective heat transfer in both the vapor and liquid phases.

We recommend that the tubes be made out of copper because it has a higher thermal diffusivity and conductivity than other realistic options, such as aluminum and other alloys.[5] Our tubing has a thickness of 5 mm in the 40 mm inner diameter section and 2 mm in the 10 mm inner diameter section. This thickness and material were selected to minimize the amount of heat stored in the tubing, maximizing transfer efficiency between the exhaust and PCM, while also maintaining the structural integrity of the tubing.

At all points in this process, the fluid flow in the tube is turbulent, though the Reynolds number does vary. For internal forced convection (as this system can be modeled), a fluid is experiencing turbulent flow if the Reynolds number exceeds 2300,[6] and our fluid flow is always above this threshold value. Logically, it is known that increasingly turbulent flow results in greater convection and more efficient heat transfer. For single-phase fluids (such as steps 1 and 3), we were able to obtain the convection coefficient using the Dittus-Boettler correlation[6] (Eq. B6) and the definition of the Nusselt Number (Eq. B7). For the convection coefficient during condensation (step 2), correlations relating pressure, mass flow rate, and tube length were utilized to determine the heat transfer flux ($q'' = 170\ kW/m^2$) for an approximately horizontal condensing steam tube[7]. We were able to correlate this heat flux and the temperature drop between steam and PCM to obtain the convection coefficient for the condensing process (Eq. B8). From each stage 1-3, a thermal resistance network can be modeled to determine the overall heat transfer coefficient (Eq. B9).

---

[4] DiPippo, R. Elements of thermodynamics, fluid mechanics, and heat transfer applied to geothermal energy conversion systems. In *Geothermal Power Generation*; DiPippo, R., Ed.; Woodhead Publishing: Kidlington, UK, 2016; pp 217–247.

[5] Wang, F.; Li, Y.; Liu, M.; Pang, D.; Du, W.; Zhang, Y.; Cheng, X.; Gu, T.; Guo, W. Comprehensive Evaluation of the Performances of Heat Exchangers with Aluminum and Copper Finned Tubes. *Int. J. Chem. Eng.* **2023**, *2023*, 1–11

[6] Goldfarb, J. *CHEME 3240: Heat and Mass Transfer—Prelim 2 Compiled Notes*; Smith School of Chemical and Biomolecular Engineering, Cornell University: Ithaca, NY, Fall 2025.

[7] Zhang, S.; Mancin, S.; Pu, L. A review and prospective of fin design to improve heat transfer performance of latent thermal energy storage. *J. Energy Storage* **2023**, *62*, 106825.

Table 2. Relevant Heat Transfer Quantities for Each Stage of Our Proposed Process

Process Stage	Prandtl Number	Reynolds Number	k (W/m*K)	Nusselt Number	h (W/m^2*K)	U (W/m^2*K)
1	1.3	1.76×10^5	0.030	401	301	299
2	1.09	1.79×10^5	0.0265	380	3063	2946
3	1.77	9.45×10^4	0.675	276	18653	15015
4	2.3	3.05×10^4	0.670	124	8318	7507
5	1.85	3.23×10^4	0.670	119	7973	7225

We model the flow in the hot coil as forced convection in a horizontal tube with a constant external fluid temperature (T_{PCM} = 78°C). The length of tubing to cool to a desired temperature (steps 1 and 3) can be determined by using the overall heat transfer coefficient and log mean temperature difference (Eq. B10). When the water in the tube is losing latent heat (step 2), a different, known correlation can be used to determine the length of the tube required to achieve our desired thermodynamic change (Eq. B11).

Table 3. Calculated Heat Transfer from the Hot Fluid and Required Tube Length

Process	Heat Transfer (kJ/s)	Length of Tubing Required (m)	Tube Diameter (mm)
1 → 2	2.97	1.25	40
2 → 3	162.15	11.38	40
3 → 4	12.33	0.80	10
4 → 5	~0	0.40	10
Total	177.45	13.83	N/A

Assuming the thermal battery is charged for 8 hours during the day (at the standard exhaust flow rate of 0.075 kg/s) and 10 hours at night (at the lower rate of 0.015 kg/s), the PCM tank would need to be capable of holding approximately 6.39 GJ (Eq. B12). Utilizing a latent heat of 280 kJ/kg[8] and assuming a constant density of solid PCM, we determined a need for 8.148 m^3 of PCM (Eq. B1). Leaving room for slight fluctuations in flow rate and the volume of our tubes, we propose a tank with a volume of 8.5 m^3.

[8] Cabeza, L. F.; Castell, A.; Barreneche, C.; De Gracia, A.; Fernández, A. I. Materials used as PCM in thermal energy storage in buildings: A review. *Renew. Sustainable Energy Rev.* **2011**, *15*, 1675–1695.

The optimal shape for a PCM tank containing coils for fluid transfer is a hexagonal shaft.[9] The dimensions of our hexagonal shaft, designed to hold 8.5 m^3 of PCM, include a base with a 1.28 m side length and a height of 2.2 m. The inner coil, containing the hot fluid, has both a turn radius and pitch of 50 cm (Figure 2), allowing for 4.3 total turns inside the heat exchanger. It is also advantageous to cover the hexagonal heat exchanger in some form of thermal insulation, ideally enough to reach the critical radius of insulation, to minimize heat loss to the surrounding environment.

A second coil can be added to our design for the flow of cold water, providing it with a different pathway from the hot exhaust stream and allowing them both to be on simultaneously, if need be. Modeling this tube similarly, the inner diameter was set to 40 mm, and the inlet temperature was assumed to be 25°C. The flow rate was set to 19.5 gal/min (1.2 kg/s), as our HX3 analysis indicated that this was the optimal flow rate for overall heat transfer. To determine the temperature change, we used this overall heat transfer coefficient and assumed a constant external fluid temperature (Eq. B13). These calculations indicate that the implementation of a separate coil, with a turn radius of 90 cm and pitch of 25 cm, which undergoes 8.6 rotations in the heat exchanger, allows for the heating of the inlet cold water to 61.5°C. The designed thermal battery is capable of this thermal transfer, assuming 7.4 hours of discharging each day, as indicated previously (Eq. B12). 3D and 2D CAD visualizations of our model are provided below in Figure 2.

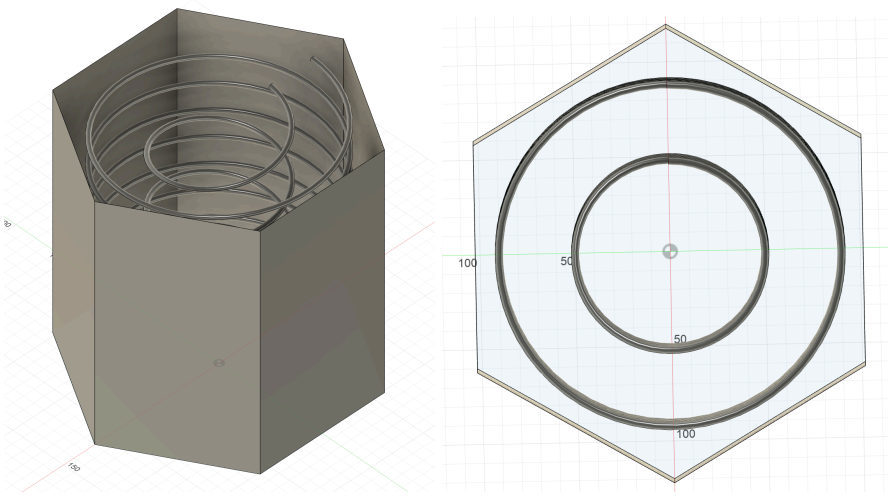
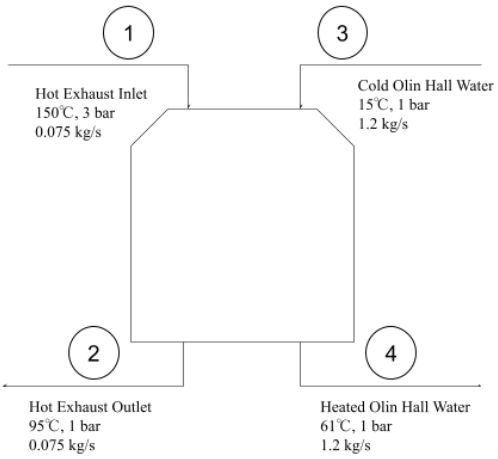


Figure 2. 3D CAD Visualization of Proposed Heat Exchanger Design with Inner and Outer Coils (left) and a 2D Top-Down View of the System (right)

If we were to utilize the same coil as the hot exhaust, using a valve system, our system would only be capable of charging or discharging at a given time. Utilizing solely the outer coil (50 cm turn radius, 50

[9] Andrzejczyk, R.; Muszynski, T.; Kowalczyk, T.; Saqib, M. Experimental and numerical investigation on shell and coil storage unit with biodegradable PCM for modular thermal battery applications. *Int. J. Therm. Sci.* **2022**, *185*, 108076. https://doi.org/10.1016/j.ijthermalsci.2022.108076.

cm pitch, 4.3 rotations) solely, the thermal battery can heat the cold stream to only 34°C (Eq. B13). The introduction of a second coil was determined to be highly worthwhile, enabling far more efficient discharging, more viable heating of Olin Hall, and a relatively small increase in financial investment. Overall results of our suggested thermal battery are summarized in Figure 3.



Process	Heat Transfer Rate (kJ/s)	Total Heat Transfer (kJ/day)
1→2	177	6.39×10^6
3→4	–239	-6.39×10^6

Figure 3. Model of Proposed Heat Exchanger Design as Single Process Unit (left) and Heat Transfer Quantities per Day (right).

INTEGRATION WITH CLIMATE ACTION PLAN

Our thermal battery design contributes to Cornell's Climate Action Plan by reducing the amount of new steam required to heat Olin Hall. Based on our system specifications, each charge stores 1.78 MWh of usable thermal energy that is utilized to heat the building (Eq. C1). Using emissions data from Cornell's 2024 Central Energy Plant report, an emission factor of 0.15 kg CO_2 per kWh of steam energy produced was applied[10] (Eq. C2). Each charge/discharge cycle avoids approximately 266.25 kg of CO_2 (Eq. C3), and assuming one full cycle per day during the heating season (~250 days), the system would reduce emissions by 66,563 kg CO_2 per year (Eq. C4).

This design directly supports the "reduce" and "replace" tiers of Cornell's carbon neutrality hierarchy by prioritizing energy conservation and efficiency improvements before large-scale infrastructure can be implemented. The proposed thermal battery reduces steam demand at the building level, allowing existing infrastructure to operate more efficiently while lowering overall emissions.
Currently, Cornell is investigating Earth Source Heat (ESH) as a long-term strategy to replace the steam line entirely. ESH would draw on geothermal heat deep below the surface to supply a renewable,

[10] Cornell Energy and Sustainability. *Fiscal Year 2024 Cornell University Central Energy Plant (CEP) Fast Facts*; Cornell University: Ithaca, NY, 2025. https://fcs.cornell.edu/sites/default/files/2025-03/FY-2024_Cornell%20Fast%20Facts.pdf (accessed December 16, 2025).

high-temperature source of hot water to campus buildings.[11] While exploratory drilling has confirmed that the subsurface temperatures are sufficient for this approach, further work is required to determine whether the rock can deliver heat at usable flow rates.[12] Because ESH will take additional time to access, design, and construct, our thermal battery serves as an immediate solution; it supports the transition by addressing needs that exist before geothermal heat is available. Building level storage would still be valuable under ESH because it helps balance a steady geothermal supply with the building's changing demand.

RECOMMENDATIONS AND CONCLUSIONS

Based on the aforementioned analyses, we recommend that our thermal battery design be incorporated into Olin Hall. Our hexagonal tank, filled with barium hydroxide octahydrate PCM, serves as a very effective medium of heat storage, as it is capable of storing the 6.39 GJ of energy transferred from the hot exhaust stream each day in the form of latent heat. We encourage allowing the hot stream to fully condense and cool to 95°C to maximize heat transfer into the battery, and the cold stream at approximately 4.2 m/s. To maximize energy extraction from the hot stream, we encourage adjusting the tube diameter to achieve the desired flow rate for vapor and liquid states; going from a 40 mm to a 10 mm inner diameter accomplishes this.

Our two-coil setup allows for more heat to be transferred to the PCM per unit volume, as a longer length, and consequently more surface area, of the coil is surrounded by PCM. Due to our coil arrangement, the thermal battery is only 2.2 m tall, and it requires less than 5 m^2 of floor space. This sizing is supported by appropriate heat transfer correlations and latent heat calculations for the PCM, assuring that the battery can easily fit in Olin Hall, likely even in the UO Lab or an underutilized closet. Incorporating our battery is estimated to cost upwards of 27,000 USD (considering the cost of PCM, the tubes, and installation) (Eq. D5), but we argue that this added financial burden is worth it due to the upsides of the battery. We estimate that the use of this battery will decrease Olin Hall's carbon footprint by over 66,000 kg/yr, all the while providing a significant amount of extra energy for use throughout campus.

APPENDICES

APPENDIX A: HX3 CALCULATIONS

To find U, we first found the heat transfer rate, Q, using the following known equation:

$$Q = m_h c_p (T_{h,in} - T_{h,out}) = m_c c_p (T_{c,out} - T_{c,in}) \quad \text{(Eq. A1)}$$

[11] Cornell University. *Earth Source Heat*; Cornell University: Ithaca, NY. https://earthsourceheat.cornell.edu (accessed Dec 16, 2025)

[12] Arson, C.; Balog-Way, D.; Beckers, K.; Bezner-Kerr, W.; Carson, S.; Edwards, S.; Fulton, P.; Gillenwater, M.; Goetze, T.; Gustafson, O.; Ingraffea, T.; Jordan, T.; McComas, K.; Olmstead, S.; Saltiel, S.; Tester, J.; Tucker, C.; Wells, M. *The role of Enhanced Geothermal Systems in the energy transition at Cornell. Report of a workshop held at Cornell University, Ithaca, October 23–24, 2024. arXiv* 2024, *arXiv:2412.11421*.

Using the rate of heat transfer just calculated, we used the following equation to calculate the overall heat transfer coefficient:

$$U = \frac{Q}{A \cdot \Delta T_{lm}} \quad \text{(Eq. A2)}$$

Where A is the surface area of HX3 (1.3 ft^2), and ΔT_{lm} is the log mean temperature difference:

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{ln(\Delta T_1 / \Delta T_2)} \quad \text{(Eq. A3)}$$

To determine the inner convective heat transfer coefficient, h_i, Sieder-Tate modifications were applied to the Wilson method, which relates the Nusselt number (Nu) to the Reynolds number (Re) and Prandtl number (Pr) through a viscosity ratio ϕ_v:

$$Nu = \frac{h_i D}{k} = 0.23 Re^{0.8} Pr^{1/3} \phi_v \quad \text{(Eq. A4)}$$

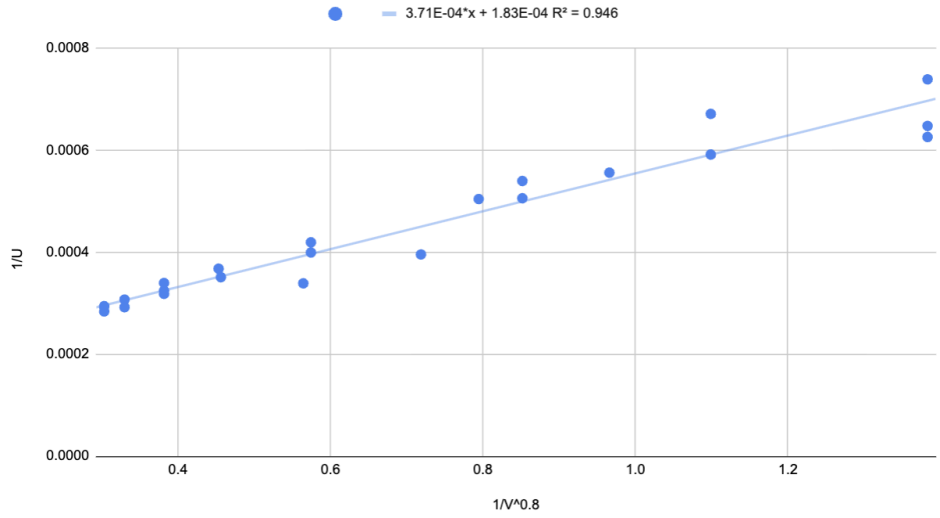
The flow regime of water within the inner tube and shell was analyzed via the Reynolds number for all data points. We observed that the Reynolds number ranged from 3,835 to 23,010, indicating solely turbulent flow. We also assume the viscosity ratio term of the two fluids is negligible for water to yield a modified Wilson plot with the following linearized equation:

$$\frac{1}{U} = m_i \left(\frac{1}{V_i^{0.8}}\right) + b_1 \quad \text{(Eq. A5)}$$

The inner convective heat transfer coefficient is equal to:

$$h_i = \frac{V_i^{0.8}}{m_i} \quad \text{(Eq. A6)}$$

Where V_i is the velocity of the hot stream. We calculated this by dividing the volumetric flow rate by the cross-sectional area, which we obtained using the diameters of the tube and shell sides found from the Olin UO Lab HX Information Packet. To graph the Wilson Plot with Sieder-Tate Correlation based on Equation A5 $\frac{1}{V_i^{0.8}}$ was plotted versus $\frac{1}{U}$. The graph yielded is as follows:



Graph 1. Wilson Plot with Sieder-Tate Correlation (1/U vs. $1/V^{0.8}$)

The slope of the line of best fit, m_i, is seen to be 3.71×10^{-4}. Equation A6 was used to calculate h_i.

To determine h_o, Donohue modifications were applied to the Wilson method:

$$Nu = \frac{h_o D_o}{k} = 0.2 Re^{0.6} Pr^{0.33} \phi_v^{0.14} \quad \text{(Eq. A7)}$$

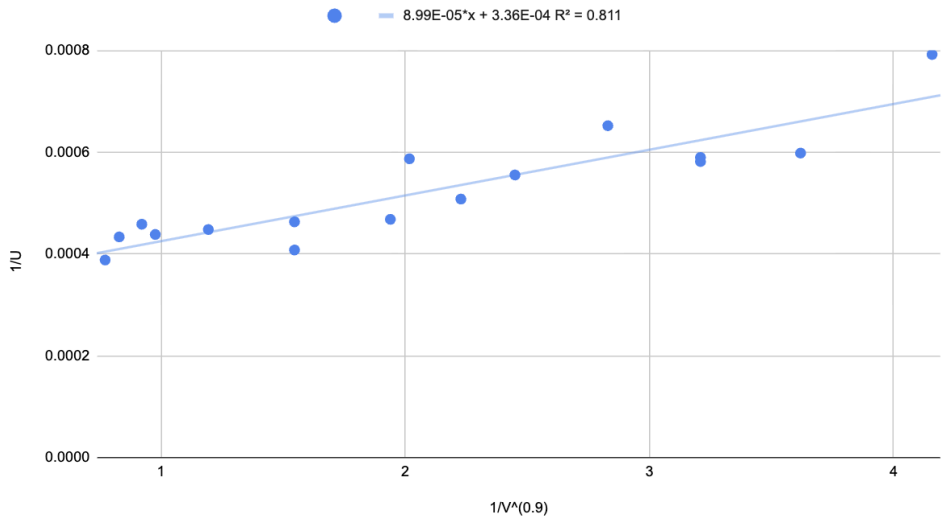
We assume that the flow in the heat exchanger is turbulent, as shown in the section above with Re > 2000. We also again assume the viscosity ratio term of the two fluids is negligible for water to yield a modified Wilson plot with the following linearized equation:

$$\frac{1}{U} = m_o\left(\frac{1}{V_o^n}\right) + b_0 \quad \text{(Eq. A8)}$$

Where the outer convective heat transfer coefficient is equal to:

$$h_o = \frac{V_o^n}{m_o} \quad \text{(Eq. A9)}$$

Where V_o is the velocity of the cold stream. We calculated this by dividing the volumetric flow rate by the cross-sectional area, which we again obtained using the diameters of the tube and shell sides. We iteratively determined *n* to equal 0.9. To graph the Wilson Plot with Donohue Correlation based on Equation A8 $\frac{1}{V_i^{0.9}}$ was plotted versus $\frac{1}{U}$. The graph is as follows:



Graph 2. Wilson Plot with Donohue Correlation (1/U vs. $1/V^{0.9}$)

The slope of the line of best fit, m_o, is 8.99×10^{-5}. Equation A9 was used to calculate the inner convective heat transfer coefficient.

APPENDIX B: THERMAL BATTERY CALCULATIONS

The mass (m) and volume (V) of PCM barium hydroxide octahydrate required to hold Q_{total}, 6.4 GJ, were determined assuming the operating temperature, a density of 2180 kg/m^3, and a latent heat of 280 kJ/kg.[1]

$$6,400,000\ kJ\left(\frac{1\ kg}{280\ kJ}\right) = 22,900\ kg\left(\frac{1\ m^3}{2180\ kg}\right) = 8.37\ m^3 \quad \text{(Eq. B1)}$$

The mass (m) and volume (V) of brick required to hold Q_{total} were determined assuming a temperature increase of 500K, a density of 700 kg/m^3, and a specific heat of 0.837 kJ/kg.[13]

$$m_{brick} = \frac{Q_{total}}{C_{p,brick}\Delta T} = 6,400,000\ kJ\left(\frac{1\ kg\ K}{0.84\ kJ}\right)\left(\frac{1}{500K}\right) = 15,238\ kg,\ V_{brick} = m_{brick}\left(\frac{1\ m^3}{700\ kg}\right) = 21.8\ m^3 \quad \text{(Eq. B2)}$$

The mass (m) and volume (V) of water required to hold Q_{total} were determined assuming a temperature increase of 50K, a density of 980 kg/m^3, and a specific heat of 4.19 kJ/kg.[14]

$$m_{water} = \frac{Q_{total}}{C_{p,water}\Delta T} = 6,400,000\ kJ\left(\frac{1\ kg\ K}{4.19\ kJ}\right)\left(\frac{1}{50K}\right) = 31,400\ kg,\ V_{water} = m_{water}\left(\frac{1\ m^3}{980\ kg}\right) = 23.36\ m^3 \quad \text{(Eq. B3)}$$

---

[13] Nick. Wood Stove Thermal Mass. Tiny Wood Stove. https://www.tinywoodstove.com/wood-stove-thermal-mass/ (accessed December 12, 2025).

[14] Barium hydroxide octahydrate. ChemicalBook. https://www.chemicalbook.com/ChemicalProductProperty_EN_CB0697590.htm (accessed December 12, 2025).

Utilizing the mass flow rate of the exhaust steam, surface area of a 40 mm and 10 mm diameter pipe, and the density of water at the temperature, pressure of the indicated state, velocity was determined. A 40 mm pipe was required to bring the velocity below 40 m/s.

$$v_{1,\,40\,mm} = \frac{\widehat{m}}{\pi r^2 \rho_1} = \frac{0.075\ kg/s}{\pi(\frac{0.040\ m}{2})^2 * 2.47 kg/m^3} = 24.16\ \text{m/s} \quad \text{(Eq. B4)}$$

$$v_{1,\,10\,mm} = \frac{\widehat{m}}{\pi r^2 \rho_1} = \frac{0.075\ kg/s}{\pi(\frac{0.010\ m}{2})^2 * 2.47 kg/m^3} = 386.61\ m/s \quad \text{(Eq. B4.1)}$$

Utilizing the mass flow rate of the exhaust steam, surface area of a 40 mm and 10 mm diameter pipe, and the density of water at the temperature, pressure of the indicated state, velocity was determined. A 10 mm pipe was required to bring the velocity to a reasonable ~ 1 m/s for liquid water.

$$v_{3,\,40\,mm} = \frac{\widehat{m}}{\pi r^2 \rho_1} = \frac{0.075\ kg/s}{\pi(\frac{0.040\ m}{2})^2 * 899 kg/m^3} = 0.062\ m/s \quad \text{(Eq. B5)}$$

$$v_{3,\,10\,mm} = \frac{\widehat{m}}{\pi r^2 \rho_1} = \frac{0.075\ kg/s}{\pi(\frac{0.010\ m}{2})^2 * 899 kg/m^3} = 1.074\ m/s \quad \text{(Eq. B5.1)}$$

The Dittus-Boettler correlation was utilized to correlate the calculated Reynolds Number (Re) and Prandtl Number (Pr) to the Nusselt Number (Nu) for each distinct state.

$$Nu = 0.023 Re^{0.8} Pr^{0.4} \quad \text{(Eq. B6)}$$

The convective heat transfer coefficient (h_i) was found via the definition of Nu, correlating to the fluid's thermal conductivity (k) and tube diameter (D).

$$h_i = \frac{Nu * k}{D} \quad \text{(Eq. B7)}$$

The convective heat transfer coefficient (h_i) was found for steps 1 and 3 in relation to the estimated heat flux (q") and temperature gradient across the pipe.

$$h_i = \frac{q"}{T_{fluid} - T_{PCM}} \quad \text{(Eq. B8)}$$

The overall convective heat transfer (U) was found in terms of the convection heat transfer coefficient (h_i), the thickness of the tube (t), and the thermal conductivity of copper (k).

$$\frac{1}{U} = \frac{1}{h_i} + \frac{t}{k} \quad \text{(Eq. B9)}$$

The length of the tube is correlated to the overall heat transfer coefficient (U), surface area (A), mass flow rate (m), perimeter (P), and specific heat (C_p). This was used for steps 1 and 3.

$$l = \frac{\widehat{m} C_{p_i}}{P\,U} ln\left(\frac{T_{in} - T_{PCM}}{T(l) - T_{PCM}}\right) \quad \text{(Eq. B10)}$$

The length of the tube is correlated to the latent heat of vaporization of water, the perimeter (P), mass flow rate (m), and the temperature difference between fluid and PCM

$$l = \frac{\widehat{m} \Delta h_{vap}}{U * P * \Delta T} \quad \text{(Eq. B11)}$$

The total charging energy of the battery was determined using the heat transfer rate of 177 kJ/s for 0.075 kg/s for 8 hours and 0.015 kg/s for 10 hours.

$$Energy\ Storage = 10\ h\left(\frac{3600\ s}{1\ h}\right)\left(\frac{177\ kJ}{s}\right)(0.20) + 8\ h\left(\frac{3600\ s}{1\ h}\right)\left(\frac{177\ kJ}{s}\right)(1.0) = 6.39\ GJ \quad \text{(Eq. B12)}$$

Utilizing log mean temperature for a fluid undergoing convection in a tube of constant external fluid temperature (PCM), we are able to determine the temperature difference as a result of heat transfer. This accounts for a tube of 48.6 m and 13.5 m, demonstrating a significant drop in efficiency.

$$T(l) = T_{PCM} + (T_{in} - T_{PCM})\ exp\left(\frac{UPl}{\widehat{m} C_p}\right) \quad \text{(Eq. B13)}$$

$$T(48.6\ m) = 78°C - (53°C)\ exp\left(\frac{(2377\ W/m^2K)(0.059\ m)(48.6\ m)}{(4.18\ kJ/kgK)(1.23\ kg/s)}\right) = 61.5°C$$

$$T(13.5\,m) \;=\; 78°C \;-\; (53°C)\,exp\left(\frac{(2377\,W/m^2K)(0.059\,m)(13.5\,m)}{(4.18\,kJ/kgK)(1.23\,kg/s)}\right) \;=\; 34.0°C$$

We determine the heat transfer rate (q) for each step of the proposed process within the hot tube. The heat extracted by step 1 was 1.7% of the total heat extracted.

$$q_1 = mC_p\Delta T_{1->2} \;=\; (0.075\,kg/s)(2.40\,kJ/kgK)(16.5K) \;=\; 2.97\,kJ/s \qquad \text{(Eq. B14)}$$

$$q_2 = m\,\Delta h_{2->3} \;=\; (0.075\,kg/s)(2162\,kJ/kg) \;=\; 162.15\,kJ/s \qquad \text{(Eq. B14.1)}$$

$$q_1 = mC_p\Delta T_{2->3} \;=\; (0.075\,kg/s)(4.27\,kJ/kgK)(38.5K) \;=\; 12.32\,kJ/s \qquad \text{(Eq. B14.2)}$$

APPENDIX C: CLIMATE ACTION PLAN CALCULATIONS

The Energy Storage in a single day (24-hour period) is converted to the energy unit kWh.

$$Energy\ Storage \;=\; 6.39\,GJ\left(\frac{277.8\,kWh}{1\,GJ}\right) = 1780\,kWh \qquad \text{(Eq. C1)}$$

The emission factor for steam generation was calculated utilizing data from Cornell's 2024 Central Energy Plant report, where M_{CO2} is the annual carbon dioxide emissions, and E_{steam} is the corresponding annual steam energy production.[10]

$$\text{EF} = \frac{M_{CO2}}{E_{steam}} = \frac{115{,}000\,metric\ tons\ CO_2}{766{,}914{,}000\,kWh} = 0.00015\,\frac{metric\ tons\ CO_2}{kWh} \;=\; 0.15\,\frac{kg\ CO_2}{kWh} \qquad \text{(Eq. C2)}$$

The carbon dioxide emissions avoided per cycle were determined using the usable thermal energy transferred during one charge/discharge cycle, Q_{stored}, and the previously calculated emission factor.

$$CO_2 \text{ saved} = Q_{stored} \cdot \text{EF} = 1775 \text{ kWh} \cdot 0.15 \text{ kg / kWh} = 266.25 \text{ kg } CO_2 \text{ per cycle} \qquad \text{(Eq. C3)}$$

Annual emission reductions were obtained assuming one charge/discharge cycle per day for the heating season, assuming 250 days in the season.

$$\text{Annual } CO_2 = CO_2 \text{ saved} \cdot N_{cycles} = 266.25 \text{ kg} \cdot 250 \text{ cycles/year} = 6.66\text{x}10^4 \text{ kg/year} \qquad \text{(Eq. C4)}$$

APPENDIX D: COST ANALYSIS

The installation cost for 2,280 kg of Barium Hydroxide Octahydrate was determined utilizing an industrial-grade listing.[2]

$$PCM\ Cost \;=\; 22{,}900\,kg\left(\frac{1\,ton}{907\,kg}\right)\left(\frac{500\,USD}{1\,ton}\right) = 12{,}624\,USD \qquad \text{(Eq. D1)}$$

The cost of installing two distinct coils[15] was determined utilizing a per-foot rate of soft copper tubing that can be easily oriented in a coil.

$$Inner\ Coil\ Cost \;=\; 13.8m\left(\frac{3.281\,ft}{1\,m}\right)\left(\frac{40\,USD}{1\,ft}\right) = 1{,}810\,USD \qquad \text{(Eq. D2)}$$

$$Outer\ Coil\ Cost \;=\; 61.4m\left(\frac{3.281\,ft}{1\,m}\right)\left(\frac{40\,USD}{1\,ft}\right) = 8{,}058\,USD \qquad \text{(Eq. D3)}$$

This is the price of eight sheets of custom-size 70 in x 80 in steel A36 plates[16] that can be cut and welded (by a Cornell Employee) to the adequate dimensions for our hexagonal shaft.

$$Hexagonal\ Shaft\ Cost \;=\; 8\,steel\,plates\left(\frac{607\,USD}{1\,steel\,plate}\right) = 4{,}856\,USD \qquad \text{(Eq. D4)}$$

$$Total\ Cost \;=\; PCM\ Cost \;+\; Inner\ Coil\ Cost \;+\; Outer\ Coil\ Cost \;+\; Hexagonal\ Shaft\ Cost \qquad \text{(Eq. D5)}$$

$$Total\ Cost \;=\; 27{,}384\,USD$$

---

[15] Mueller Industries CKS60L. Southern Pipe & Supply. https://www.southernpipe.com/7019606/Product/Mueller_Industries_CKS60L (accessed Dec 17, 2025).

[16] Steel Plate. Metals Depot. https://www.metalsdepot.com/steel-products/steel-plate (accessed Dec 17, 2025).

REFERENCES

1. Koretsky, M. D. Engineering and Chemical Thermodynamics, 2nd ed.; John Wiley: S.L., 2013.
2. Martínez, F. R.; Borri, E.; Kala, S. M.; Ushak, S.; Cabeza, L. F. Phase change materials for thermal energy storage in industrial applications. *Heliyon* **2024**, *11*, e41025. https://doi.org/10.1016/j.heliyon.2024.e41025.
3. Barium Hydroxide Octahydrate. Made-in-China.com. https://gainer.en.made-in-china.com/product/JnBYOUKyYDRA/China-CAS-17194-00-2-Factory-Price-Barium-Hydroxide-Octahydrate-99-Ba-OH-2-8H2O.html (accessed December 12, 2025).
4. DiPippo, R. Elements of thermodynamics, fluid mechanics, and heat transfer applied to geothermal energy conversion systems. In *Geothermal Power Generation*; DiPippo, R., Ed.; Woodhead Publishing: Kidlington, UK, 2016; pp 217–247.
5. Wang, F.; Li, Y.; Liu, M.; Pang, D.; Du, W.; Zhang, Y.; Cheng, X.; Gu, T.; Guo, W. Comprehensive Evaluation of the Performances of Heat Exchangers with Aluminum and Copper Finned Tubes. *Int. J. Chem. Eng.* **2023**, *2023*, 1–11. https://doi.org/10.1155/2023/6666947.
6. Goldfarb, J. *CHEME 3 240: Heat and Mass Transfer—Prelim 2 Compiled Notes*; Smith School of Chemical and Biomolecular Engineering, Cornell University: Ithaca, NY, Fall 2025.
7. Zhang, S.; Mancin, S.; Pu, L. A review and prospective of fin design to improve heat transfer performance of latent thermal energy storage. *J. Energy Storage* **2023**, *62*, 106825. https://doi.org/10.1016/j.est.2023.106825.
8. Cabeza, L. F.; Castell, A.; Barreneche, C.; De Gracia, A.; Fernández, A. I. Materials used as PCM in thermal energy storage in buildings: A review. *Renew. Sustainable Energy Rev.* **2011**, *15*, 1675–1695. https://doi.org/10.1016/j.rser.2010.11.018.
9. Andrzejczyk, R.; Muszynski, T.; Kowalczyk, T.; Saqib, M. Experimental and numerical investigation on shell and coil storage unit with biodegradable PCM for modular thermal battery applications. *Int. J. Therm. Sci.* **2022**, *185*, 108076. https://doi.org/10.1016/j.ijthermalsci.2022.108076.
10. Cornell Energy and Sustainability. *Fiscal Year 2024 Cornell University Central Energy Plant (CEP) Fast Facts*; Cornell University: Ithaca, NY, 2025. https://fcs.cornell.edu/sites/default/files/2025-03/FY-2024_Cornell%20Fast%20Facts.pdf
11. Cornell University. *Earth Source Heat*; Cornell University: Ithaca, NY. https://earthsourceheat.cornell.edu (accessed Dec 16, 2025)
12. Arson, C.; Balog-Way, D.; Beckers, K.; Bezner-Kerr, W.; Carson, S.; Edwards, S.; Fulton, P.; Gillenwater, M.; Goetze, T.; Gustafson, O.; Ingraffea, T.; Jordan, T.; McComas, K.; Olmstead, S.; Saltiel, S.; Tester, J.; Tucker, C.; Wells, M. *The role of Enhanced Geothermal Systems in the energy transition at Cornell. Report of a workshop held at Cornell University, Ithaca, October 23–24, 2024. arXiv* 2024, *arXiv:2412.11421*. https://doi.org/10.48550/arXiv.2412.11421
13. Nick. Wood Stove Thermal Mass. Tiny Wood Stove. https://www.tinywoodstove.com/wood-stove-thermal-mass/ (accessed December 12, 2025).
14. Barium hydroxide octahydrate. ChemicalBook. https://www.chemicalbook.com/ChemicalProductProperty_EN_CB0697590.htm (accessed December 12, 2025).
15. Mueller Industries CKS60L. Southern Pipe & Supply. https://www.southernpipe.com/7019606/Product/Mueller_Industries_CKS60L (accessed Dec 17, 2025).
16. Steel Plate. Metals Depot. https://www.metalsdepot.com/steel-products/steel-plate (accessed Dec 17, 2025).